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Article

Reinforcement Learning in Robotics: Applications and Real-World Challenges †

Petar Kormushev *, Sylvain Calinon and Darwin G. Caldwell

Department of Advanced Robotics, Istituto Italiano di Tecnologia, via Morego 30, 16163 Genova, Italy; E-Mails: sylvain.calinon@iit.it (S.C.); darwin.caldwell@iit.it (D.G.C.)

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* Author to whom correspondence should be addressed; E-Mail: petar.kormushev@iit.it.

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Abstract: In robotics, the ultimate goal of reinforcement learning is to endow robots with the ability to learn, improve, adapt and reproduce tasks with dynamically changing constraints based on exploration and autonomous learning. We give a summary of the state-of-the-art of reinforcement learning in the context of robotics, in terms of both algorithms and policy representations. Numerous challenges faced by the policy representation in robotics are identified. Three recent examples for the application of reinforcement learning to real-world robots are described: a pancake flipping task, a bipedal walking energy minimization task and an archery-based aiming task. In all examples, a state-of-the-art expectation-maximization-based reinforcement learning is used, and different policy representations are proposed and evaluated for each task. The proposed policy representations offer viable solutions to six rarely-addressed challenges in policy representations: correlations, adaptability, multi-resolution, globality, multi-dimensionality and convergence. Both the successes and the practical difficulties encountered in these examples are discussed. Based on insights from these particular cases, conclusions are drawn about the state-of-the-art and the future perspective directions for reinforcement learning in robotics.

Keywords: reinforcement learning; robotics; learning and adaptive systems

1. Introduction

Endowing robots with human-like abilities to perform motor skills in a smooth and natural way is one of the important goals of robotics. A promising way to achieve this is by creating robots that can learn new skills by themselves, similarly to humans. However, acquiring new motor skills is not simple and involves various forms of learning.

Over the years, the approaches for teaching new skill to robots have evolved significantly, and currently, there are three well-established types of approaches: *direct programming*, *imitation learning* and *reinforcement learning*. All of these approaches are still being actively used, and each one has its own advantages and disadvantages and is preferred for certain settings, as summarized in Table 1. The bottom row in Table 1 indicates our prediction about a hypothetical future approach, predicted based on extrapolation from the existing approaches, as explained in Section 10.

The ultimate goal of these approaches is to give robots the ability to learn, improve, adapt and reproduce tasks with dynamically changing constraints. However, these approaches differ significantly from one another:

- **Direct programming**: This is the lowest-level approach, but it is still being actively used in industrial settings, where the environment is well-structured and complete control over the precise movement of the robot is crucial. We add it for completeness, although it is not really a *teaching* method; hence, we classify it as *programming*. Industrial robots are usually equipped with the so-called *teach pendant*—a device that is used to manually set the desired robot positions.
- **Imitation learning**: This approach is called *learning* instead of *programming*, in order to emphasize the active part that the agent (the robot) has in the process. This approach is also known as *programming by demonstration* [1] or *learning from demonstration* [2]. There are three main methods used to perform demonstrations for imitation learning:
  - **Kinesthetic teaching**: This is the process of manually moving the robot's body and recording its motion [3]. This approach usually works only for smaller, lightweight robots and in combination with a gravity-compensation controller, in order to minimize the apparent weight of the robot. However, nevertheless, since the robot's inertia cannot be effectively reduced, it is not practical for bigger robots. Kinesthetic teaching could be performed in a continuous way, recording whole trajectories, or alternatively, it could be performed by recording discrete snapshots of the robot's state at separate time instances, such as in keyframe-based teaching of sequences of key poses [4].
  - **Teleoperation**: This is the process of remotely controlling the robot's body using another input device, such as a joystick or a haptic device. This approach shares many similarities with kinesthetic teaching in terms of advantages and disadvantages. One big difference is that with teleoperation, the teacher can be located in a geographically distant location. However, time delays become an issue as the distance increases (e.g., teleoperating a Mars rover would be difficult). Similarly to kinesthetic teaching, only a limited number of degrees of freedom (DoFs) can be controlled at the same time. Furthermore, with teleoperation, it is more difficult to feel the limitations and capabilities of the robot than by using kinesthetic

teaching. As an advantage, sometimes the setup could be simpler, and additional information can be displayed at the interface (e.g., forces rendered by the haptic device).

- **Observational learning**: In this method, the movement is demonstrated using the teacher's own body and is perceived using motion capture systems or video cameras or other sensors. It is usually needed to solve the correspondence problem [5], *i.e.*, to map the robot's kinematics to that of the teacher.

The simplest way to use the demonstrations is to do a simple record-and-replay, which can only execute a particular instance of a task. However, using smart representation of the recorded movement in appropriate frame of reference (e.g., using the target object as the origin), it is possible to have somewhat adaptable skill to different initial configurations, for simple (mostly one-object) tasks. Based on multiple demonstrations that include variations of a task, the robot can calculate correlations and variance and figure out which part of a task is important to be repeated verbatim and which part is acceptable to be changed (and to what extent). Imitation learning has been successfully applied many times for learning tasks on robots, for which the human teacher can demonstrate a successful execution [3,6].

- **Reinforcement learning (RL)**: This is the process of learning from trial-and-error [7], by exploring the environment and the robot's own body. The goal in RL is specified by the *reward function*, which acts as positive reinforcement or negative punishment depending on the performance of the robot with respect to the desired goal. RL has created a well-defined niche for its application in robotics [8–14]. The main motivation for using reinforcement learning to teach robots new skills is that it offers three previously missing abilities:
  - to learn new tasks, which even the human teacher cannot physically demonstrate or cannot directly program (e.g., jump three meters high, lift heavy weights, move very fast, *etc.*);
  - to learn to achieve optimization goals of difficult problems that have no analytic formulation or no known closed form solution, when even the human teacher does not know what the optimum is, by using only a known cost function (e.g., minimize the used energy for performing a task or find the fastest gait, *etc.*);
  - to learn to adapt a skill to a new, previously unseen version of a task (e.g., learning to walk from flat ground to a slope, learning to generalize a task to new previously unseen parameter values, *etc.*). Some imitation learning approaches can also do this, but in a much more restricted way (e.g., by adjusting parameters of a learned model, without being able to change the model itself).

Reinforcement learning also offers some additional advantages. For example, it is possible to start from a "good enough" demonstration and gradually refine it. Another example would be the ability to dynamically adapt to changes in the agent itself, such as a robot adapting to hardware changes—heating up, mechanical wear, growing body parts, *etc.*

Table 1. Comparison of the main robot teaching approaches.

		Approach/method for teaching the robot	Computational complexity Difficulty for the teacher	Advantages	Disadvantages
Existing approaches		**Direct programming** Manually specifying (programming) the robot how to move each joint	Lowest High	Complete control of the movement of the robot to the lowest level	Time-consuming, error-prone, not scalable, not reusable
Existing approaches	Imitation learning	**Kinesthetic teaching** and **Teleoperation** Directly move the robot's limbs or teleoperate to record the movement	Low Medium	No need to manually program, can just record and replay movements	Cannot perform fast movements; can move usually only one limb at a time; the robot has to be lightweight
Existing approaches	Imitation learning	**Observational learning** Demonstrate the movement using the teacher's own body; perceive the movement using motion capture systems or video cameras	Medium Low	Easy and natural to demonstrate; also works for bimanual tasks or even whole-body motion	Correspondence problem caused by the different embodiment; the teacher must be able to do the task; often requires multiple demonstrations that need to be segmented and time-aligned
Existing approaches		**Reinforcement learning** Specify a scalar reward function evaluating the robot's performance that needs to be maximized; no need to demonstrate how to perform the task; the robot discovers this on its own	Higher Lower	Robot can learn tasks that even the human cannot demonstrate; novel ways to reach a goal can be discovered	No control over the actions of the robot; robot has only indirect information about the goal; need to specify reward function, policy parameterization, exploration magnitude/strategy, initial policy
Hypothetical		**"Goal-directed learning"** (predicted via extrapolation) Only specifying the goal that the robot must achieve, without evaluating the intermediate progress	Highest Lowest	The easiest way to specify (e.g., using NLP); robot has direct knowledge of the goal	No control of the movement of the robot; must know what the goal is and how to formally define it

This paper provides a summary of some of the main components for applying reinforcement learning in robotics. We present some of the most important classes of learning algorithms and classes of policies. We give a comprehensive list of challenges for effective policy representations for the application of policy-search RL to robotics and provide three examples of tasks demonstrating how the policy representation may address some of these challenges. The paper is a significantly improved and extended version of our previous work in [15]. The novelties include: new experimental section based on robotic archery task; new section with insights about the future of RL in robotics; new comparison of existing robot learning approaches and their respective advantages and disadvantages; an expanded list of challenges for the policy representation and more detailed description of use cases.

The paper does not propose new algorithmic strategies. Rather, it summarizes what our team has learned from a fairly extensive base of empirical evidence over the last 4–5 years, aiming to serve as a reference for the field of robot learning. While we may still dream of a general purpose algorithm that would allow robots to learn optimal policies without human guidance, it is likely that these are far off. The paper describes several classes of policies that have proved to work very well for a wide range of robot motor control tasks. The main contribution of this work is a better understanding that the design of appropriate policy representations is essential for RL methods to be successfully applied to real-world robots.

The paper is structured as follows. In Section 2, we present an overview of the most important recent RL algorithms that are being successfully applied in robotics. Then, in Section 3, we identify numerous challenges posed by robotics on the RL policy representation, and in Section 4, we describe the state-of-the-art policy representations. To illustrate some of these challenges, and to propose adequate solutions to them, the three consecutive Sections 5–7, give three representative examples for real-world application of RL in robotics. The examples are all based on the same RL algorithm, but each faces different policy representation problems and, therefore, requires different solutions. In Section 8, we give a summary of the three tasks and compare them to three other robot skill learning tasks. Then, in Section 9, we give insights about the future perspective directions for RL based on these examples and having in mind robotics, in particular, as the application. Finally, in Section 10, we discuss potential future alternative methods for teaching robots new tasks that might appear. We conclude with a brief peek into the future of robotics, revealing, in particular, the potential wider need for RL.

2. State-of-the-Art Reinforcement Learning Algorithms in Robotics

Robot systems are naturally of high-dimensionality, having many degrees of freedom (DoF), continuous states and actions and high noise. Because of this, traditional RL approaches based on MDP/POMDP/discretized state and action spaces have problems scaling up to work in robotics, because they suffer severely from the curse of dimensionality. The first partial successes in applying RL to robotics came with the function approximation techniques, but the real "renaissance" came with the policy-search RL methods.

In policy-search RL, instead of working in the huge state/action spaces, a smaller policy space is used, which contains all possible policies representable with a certain choice of policy parameterization. Thus, the dimensionality is drastically reduced and the convergence speed is increased.

Until recently, *policy-gradient algorithms* (such as Episodic Natural Actor-Critic eNAC [16] and Episodic REINFORCE [17]) have been a well-established approach for implementing policy-search RL [10]. Unfortunately, policy-gradient algorithms have certain shortcomings, such as high sensitivity to the learning rate and exploratory variance.

An alternative approach that has gained popularity recently derives from the Expectation-Maximization (EM) algorithm. For example, Kober *et al*. proposed in [18] an episodic RL algorithm, called PoWER (*policy learning by weighting exploration with the returns*). It is based on the EM algorithm and, thus, has a major advantage over policy-gradient-based approaches: it does not require a learning rate parameter. This is desirable, because tuning a learning rate is usually difficult to do for control problems, but critical for achieving good performance of policy-gradient algorithms. PoWER also demonstrates superior performance in tasks learned directly on a real robot, by applying an importance sampling technique to reuse previous experience efficiently.

Another state-of-the-art policy-search RL algorithm, called *PIˆ2* (*policy improvement with path integrals*), was proposed by Theodorou *et al.* in [19], for learning parameterized control policies based on the framework of stochastic optimal control with path integrals. They derived update equations for learning to avoid numerical instabilities, because neither matrix inversions nor gradient learning rates are required. The approach demonstrates significant performance improvements over gradient-based policy learning and scalability to high-dimensional control problems, such as control of a quadruped robot dog.

Several search algorithms from the field of stochastic optimization have recently found successful use for iterative policy improvement. Examples of such approaches are the cross-entropy method (CEM) [20] and the covariance matrix adaptation evolution strategy (CMA-ES) [21]. Although these algorithms come from a different domain and are not well-established in RL research, they seem to be a viable alternative for direct policy search RL, as some recent findings suggest [22].

3. Challenges for the Policy Representation in Robotics

Only having a good policy-search RL algorithm is not enough for solving real-world problems in robotics. Before any given RL algorithm can be applied to learn a task on a robot, an appropriate *policy representation* (also called *policy encoding*) needs to be devised. This is important, because the choice of policy representation determines what in principle can be learned by the RL algorithm (*i.e.*, the policy search space), analogous to the way a hypothesis model determines what kind of data a regression method can fit well. In addition, the policy representation can have significant influence on the RL algorithm itself, e.g., it can help or impede the convergence or influence the variance of the generated policies.

However, creating a good policy representation is not a trivial problem, due to a number of serious *challenges* posed by the high requirements from a robotic system, such as:

- *smoothness*—the policy representation needs to encode smooth, continuous trajectories, without sudden accelerations or jerks, in order to be safe for the robot itself and also to reduce its energy consumption. In some rare cases, though, such as in bang-bang control, sudden changes might be desirable;
- *safety*—the policy should be safe, not only for the robot (in terms of joint limits, torque limits, work space restrictions, obstacles, *etc.*), but also for the people around it;
- *gradual exploration*—the representation should allow gradual, incremental exploration, so that the policy does not suddenly change by a lot; e.g., in state-action-based policies, changing the policy action at only a single state could cause a sudden dramatic change in the overall behavior of the system when following this new branch of the policy, which is not desirable neither for the robot, nor for the people around it. In some cases, though, a considerable step change might be necessary, e.g., a navigation task where a robot needs to decide whether to go left or right to avoid an obstacle;
- *scalability*—to be able to scale up to high dimensions and for more complex tasks; e.g., a typical humanoid robot nowadays has well above 50 DoF;

- *compactness*—despite the high DoF of robots, the policy should use very compact encoding, e.g., it is impossible to directly use all points on a trajectory as policy parameters;
- *adaptability*—the policy parameterization should be adaptable to the complexity and fidelity of the task, e.g., lifting weights *vs.* micro-surgery;
- *multi-resolution*—different parts of the policy parameterization should allow different resolution/precision;
- *unbiasedness*—the policy parameterization should work without prior knowledge about the solution being sought and without restricting unnecessarily the search scope for possible solutions;
- *prior/bias*—whenever feasible, it should be possible to add prior information (also called *bias*) in order to jump-start policy search approaches, as illustrated in some of the use cases in this paper;
- *regularization*—the policy should allow one to incorporate regularization to guide the exploration towards desired types of policies;
- *time-independence*—this is the property of a policy not to depend on precise time or position, in order to cope with unforeseen perturbations;
- *embodiment-agnostic*—the representation should not depend on any particular embodiment of the robot, e.g., joint-trajectory based policies cannot be transferred to another robot easily;
- *invariance*—the policy should be an invariant representation of the task (e.g., rotation-invariant, scale-invariant, position-invariant, *etc.*);
- *correlations*—the policy should encapsulate correlations between the control variables (e.g., actuator control signals), similar to the motor synergies found in animals;
- *globality*—the representation should help the RL algorithm to avoid local minima;
- *periodicity*—to be able to represent easily periodic/cyclic movements, which occur often in robotics (e.g., for locomotion—different walking gaits, for manipulation—wiping movements, *etc.*);
- *analyzability*—facility to visualize and analyze the policy (e.g., proving its stability by poles analysis, *etc.*);
- *multi-dimensionality*—to be able to use efficiently high-dimensional feedback without the need to convert it into a scalar value by using a weighted sum of components;
- *convergence*—to help the RL algorithm to converge faster to the (possibly local) optima.

A good policy representation should provide solutions to all of these challenges. However, it is not easy to come up with such a policy representation that satisfies all of them. In fact, the existing state-of-the-art policy representations in robotics cover only subsets of these requirements, as highlighted in the next section.

4. State-of-the-Art Policy Representations in Robotics

Traditionally, explicit time-dependent approaches, such as cubic splines or higher-order polynomials, were used as policy representations. These, however, are not autonomous, in the sense that they cannot cope easily with perturbations (unexpected changes in the environment). Currently, there are a number of efficient state-of-the-art representations available to address this and many of the other challenges mentioned earlier. We give three examples of such policy representations below: